A comparison of Naive Bayes (NB) and Logistic Regression (LR) on predicting whether an individual has heart disease or not.

Description and motivation:

- Solve the binary classification problem of predicting whether someone has heart disease based on certain attributes.
- These Attributes include Age, Sex, Chest Pain, Trestbps , Chol , FBS, Resting, restecg, Thalach, Exang, Oldpeak, Slope, Ca & Thal
- A heart attack (Cardiovascular diseases) occurs when the flow of blood to the heart muscle suddenly becomes blocked. 17.9 million dying from heart attacks a year.
- Compare and analyse performance of the two models, using performance measure to analyse and evaluate findings.
- Data-set does not have significant biased to a particular output as seen in figure 2.
- Will be referring to papers [1],[2] &[3] for guidance, even though the Dataset used were different, it will give a good all-round understanding.

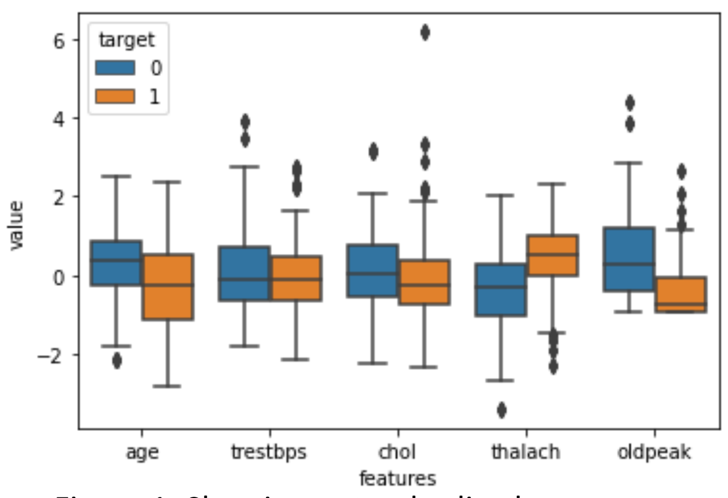

Figure 1: Showing a standardized version of the continuous features in dataset

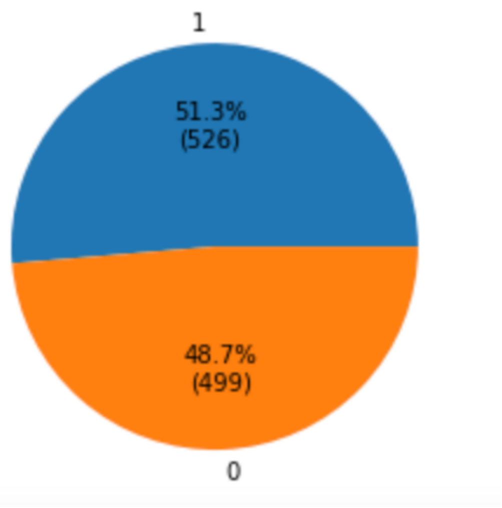

Figure 2: Showing the % values of people who have heart disease or not.

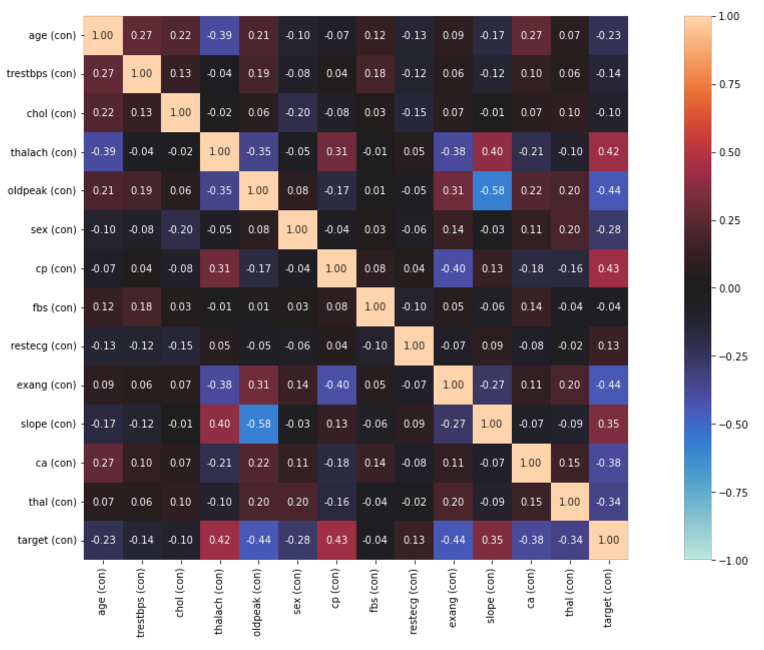

Figure 3: Showing the matrix correlation of all the features.

Hypothesis Statement

- Papers we have is based on different data & NB & LR expected to both perform well.
- If conclusive, i.e., produce similar results able to provide medical advice and attention to those.
- Papers I have read [1-7] NB-84.25% to 86.6%, LR- 82.56% to 84.85% expect to see the same.

Training and Evaluation Methodology

- Date set will be split into train (80%) and test set (20**%).**
- A baseline model for each model is created, with training Data. Then be binned and feature reduction applied (Train Set).
- Choses features are "chol","thalach",'oldpeak','sex','cp','restecg','exang','slope',"binAge",'ca','thal','target'. Age was binned to form 'binAge'. Highly correlated feat remov
- Respective hyperparameters are applied to the models and will be validated by K-Fold validation (10)
- 10-Fold validation used to estimate the generalisation error, prevents overfitting.
- Apply optimization to choose the optimum parameter and validated using 10-Fold validation.
- Testing the two chosen models with their best parameters on the testing data and contrasting the results to each other.

Choice of parameters and experimental results

Naïve Bayes :

- Classification algorithm based on Bayes Theorem
- Assumes independence amongst features.
- Takes a set of probabilities of each feature given its class ('likelihood') and multiplied class probabilities('priors')
- MAP decision rule assigns observation predicted class.

Pros:
- Easy to implement
- Requires a small amount of training to estimate test.

Cons:
- Independent assumption not always met.
- Outperformed by more sophisticated models.
- Works better with continuous features than numerical

Table 1: Showing the training accuracy of all the features

Training Model	ACCURACY TRANING %
NB-Base	83.54
NB-Best	91.71
LR-Base	84.27
LR-Best	92.07

Parameters
- Baseline model: was improved by removing highly correlated features and redundant.
- Change Distribution type of the features (Kernel, Normal, Multinomial for respective feature
- Find the optimum combination and apply kernel width to optimize

Results:
- Gaussian distribution proceed to be the best out of the distributions.
- Kernel width optimization', being applied, it improved the model further.
- All models had similar false positive and false negative

Logistic Regression:

- Regression analysis used to predict probability of a binary outcome.
- Uses MLE to estimate parameters given that the data is under a certain type of distribution.
- Describes data to explain relationship between one dependant binary variable and one or more nominal, interval or ration 0 level independent variables.[7]

Pros:
- Efficient method which required less computation and less storage.
- Has low variance

Cons:
- Not work well with few observations.
- Less observations may lead to overfitting.

Parameters
- Model was improved by applying feature selection and taking out redundant features.
- Ridge Regularisation was applied.

Results
- Using ridge regularization and setting lambda to 0.0062, increased the performance of the model.

Lessons Learnt
- For Naïve bayes, explore looking at the prior. Changing the type of proper distribution mat improve model.
- Use more new/ relevant data/ combine datasets to see outcome.
- Don't always have to optimize model, fits the training data too well.

Future Work
- Apply other ML techniques and consider joining techniques together. Gather more data and joint them.. Getter a wide range, which possible could increase accuracy.
- Consider trying to create an imbalanced dataset.
- Use another ML model and possibly combine two models to create an optimum one.

Analysis and Critical Evaluation

- Both models have high accuracy in the training phase. The Base model was improved by applying regularization techniques , with LR model having higher accuracy in the training phase.
- Applying both models to the testing data we see that there was a decrease in accuracy, with LR decreasing quite significantly from 92.07 to 80% I and NB dropped from 91.71% to 90% in terms of accuracy. This tells us that we overfitted the training data, especially for LR and more significantly, NB outperformed LR, with higher performance, which most papers agreed with, but not the magnitude.
- This tells us that Looking into this further; we can see that 'Recall' was significantly greater for NB than LR, shows that it to detect people with heart disease efficiently, highlighting the fact that logistic regression has been over trained.
- ROC curve, figure4, shows that NB has a higher AUC than, LR. Shows that it fits dataset better than LR
- Comparing this to paper[3]; we see that there are indifference in results,[NB-90.78%, LR-92.76%], especially for LR which could be since the data used where different as well as that the way the data in terms of features were collected was also be different.
- Results for NB, where like like those results [3], even though the data is different, shows, NB performs well with different datasets.

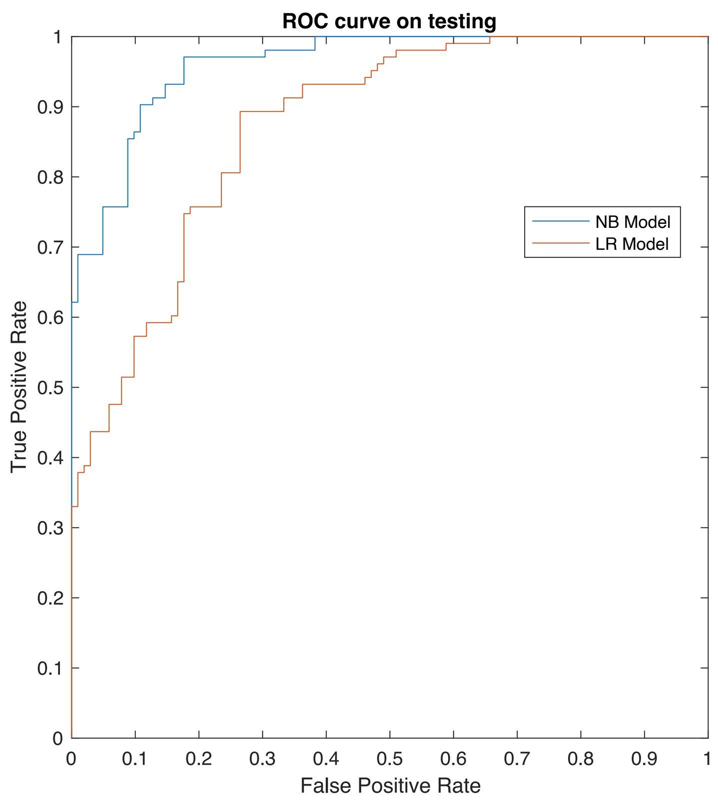

Figure 4: Showing the ROC curve for both models on the testing data.

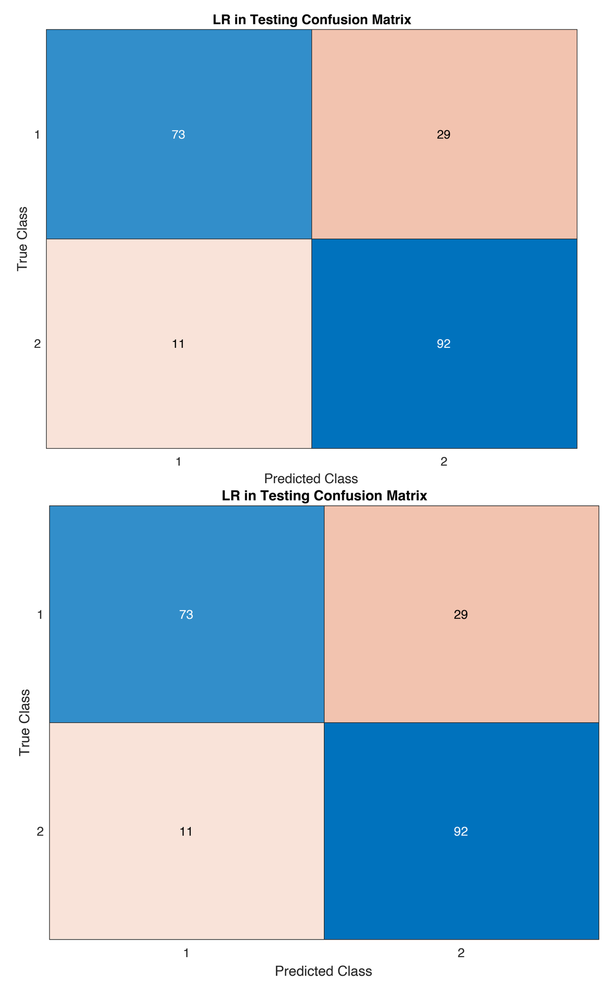

Figure 4: Showing confusion matrix of both NB and LR..

Table 2: Showing table of performance outputs of the two best models, one LR and 1 NB on the test data.

Output	NB-Testing	LR-Testing
Accuracy	90%	80%
Precision	90%	87%
Recall	89%	72%
Specificity	90%	89%
F1 Score	90%	78%

References

[1] S. Bashir, Z. S. Khan, F. Hassan Khan, A. Anjum and K. Bashir, "Improving Heart Disease Prediction Using Feature Selection Approaches," *2019 16th International Bhurban Conference on Applied Sciences and Technology (IBCAST)*, 2019, pp. 619-623, doi: 10.1109/IBCAST.2019.8667106.https://ieeexplore.ieee.org/abstract/document/8667106/citations#citations

[2] DEEPTHI, GUNTURU, KORUPROLU NAGAVINITH, and KESUBOYINA HANUDEEP -. 'HEART DISEASE PREDICTION USING MACHINE LEARNING ALGORITHMS'. *UGC AUTONOMOUS*,2021 2017, 35.

[3] S. M. M. Hasan, M. A. Mamun, M. P. Uddin and M. A. Hossain, "Comparative Analysis of Classification Approaches for Heart Disease Prediction," *2018 International Conference on Computer, Communication, Chemical, Material and Electronic Engineering (IC4ME2)*, 2018, pp. 1-4, doi: 10.1109/IC4ME2.2018.8465594. https://ieeexplore.ieee.org/abstract/document/8465594

[4]https://reader.elsevier.com/reader/sd/pii/S2666285X22000449?token=893986DA8397ECF5C7E71C6C19995AA3D356EF43166EC975330DDE92B81D525D36498FD483189DCFEA13EDF577DD1A99&originRegion=eu-west-1&originCreation=20221222212336 – 80-20

[5] Gao, L., Ding, Y. Disease prediction via Bayesian hyperparameter optimization and ensemble learning. *BMC Res Notes* **13**, 205 (2020). https://doi.org/10.1186/s13104-020-05050-0 https://d-nb.info/1212702441/34 - (Why bayes optmizaton used , it shows to perform better in Grid search and random search)

•[6] Akkaya, Berke & Çolakoğlu, Nurdan. (2019). Comparison of Multi-class Classification Algorithms on Early Diagnosis of Heart Diseases.

•[7] https://www.statisticssolutions.com/free-resources/directory-of-statistical-analyses/what-is-logistic-regression/